

## Research paper

# Association between the performance of executive function and the remission of depressive state after clinical treatment in patients with depression

Yongqing Hou<sup>a,b,c,\*</sup>, Wen Liu<sup>c</sup>, Tianbao He<sup>c</sup>, Antao Chen<sup>d,\*\*</sup>

<sup>a</sup> Department of Neurobiology and Department of Neurology of Second Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou 310058, China

<sup>b</sup> National Health Commission and Chinese Academy of Medical Sciences Key Laboratory of Medical Neurobiology, Ministry of Education Frontier Center of Brain Science and Brain-machine Integration, School of Brain Science and Brain Medicine, Zhejiang University, Hangzhou 310058, China

<sup>c</sup> Mental Health Center of Guangyuan, Sichuan, Guangyuan 628000, China

<sup>d</sup> School of Psychology, Shanghai University of Sport, Shanghai 200438, China

## ARTICLE INFO

## Keywords:

Depression

Executive function

Clinical prognosis

Depression remission

Working memory

## ABSTRACT

**Background:** Previous studies have reported that patients with depression have significant cognitive impairment. The aim of this study is to comprehensively evaluate the impairment of executive functions in patients with depression and whether the cognitive behavior performance of executive function is association with remission of depressive state after clinical treatment.

**Methods:** We used cognitive-behavioral test to evaluate the performance of executive functions of 95 inpatients with depression before hospitalization and conducted two follow-up evaluations of their depression status on the 15th day of hospitalization and approximately 9 months after discharge.

**Results:** The performance of executive function except the accuracy of inhibition control in patients with depression were significantly worse than that of healthy controls. Multivariate linear regression analysis found that the reaction time of working memory not only had a significant linear relationship with the baseline depression scores of patients with depression, but also had a significant linear relationship with the reduced depression scores after two follow-up visits.

**Limitations:** We only used cognitive-behavioral data as indicators to evaluate the cognitive performances of participants and only measured three components of executive function.

**Conclusions:** The reaction time of working memory was a stable and effective predictor of symptom relief in patients with depression after clinical treatment. These results provide initial evidence for working memory to predict the clinical prognosis of inpatients with depression prospectively, which could be further leveraged to improve intervention approaches and analyze the heterogeneity of depression.

## 1. Introduction

Depression is a common and easily occurring mental disease, which seriously limits the individual's psychosocial function and reduces people's quality of life (Malhi and Mann, 2018). According to the World Health Organization (WHO), depression has one of the highest disability rates in the World, with >300 million people worldwide suffering from it and showing a rapid increase (Liu et al., 2020; World Health Organization, 2017). In particular, during the COVID-19 epidemic, the prevalence of depression has increased significantly (Daly and

Robinson, 2022; Li et al., 2021b) and has imposed a great burden and distress on patients and their families (COVID-19 Mental Disorders Collaborators, 2021). Therefore, to explore the characteristics of patients with depression in behavioral, cognitive, and neurophysiological aspects, and identify the factors that can effectively predict the clinical treatment effect of depression, it is of great significance to understand the course of depression and to carry out the targeted treatment of depression.

Depression clinically manifests as persistent mood decline, decreased interest, and lack of energy, accompanied by significant cognitive

\* Correspondence to: Y. Hou, School of Brain Science and Brain Medicine, Zhejiang University, No. 866 Yuhangtang Road, Xihu District, Hangzhou 310058, China.

\*\* Correspondence to: A. Chen, School of Psychology, Shanghai University of Sport, No. 399 Changhai Road, Yangpu District, Shanghai 200438, China.

E-mail addresses: [houyq@zju.edu.cn](mailto:houyq@zju.edu.cn) (Y. Hou), [chenantao@sus.edu.cn](mailto:chenantao@sus.edu.cn) (A. Chen).

<https://doi.org/10.1016/j.jad.2024.07.137>

Received 27 September 2023; Received in revised form 9 July 2024; Accepted 16 July 2024

Available online 20 July 2024

0165-0327/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

decline (Malhi and Mann, 2018). Some patients may exhibit self-harm and suicidal behavior (Cummings, 2023), and may even be accompanied by psychotic symptoms such as delusions and hallucinations (Rodríguez-Pascual et al., 2021). According to the cognitive-emotional interaction theory, people's cognitive style (that is, information processing style) is closely related to the acquisition of information, and their interaction affects the formation of individual's learning and belief or cognitive schema, these beliefs or schemas affect an individual's behavior or decision-making either explicitly or implicitly (Liu et al., 2009; Rudolph et al., 2017). The long-term accumulated negative learning experience and cognitive schema are the core factors of forming harmful cognitive style, which is an important cause of inducing depression (Dehn and Beblo, 2019). The impairment of an individual's cognitive function inevitably leads to a decline in the individual's ability to acquire and process information, which in turn exacerbates the generation of adverse cognitive patterns and exacerbates an individual's depression (Knight and Baune, 2018). Cognitive impairment is also an important cause of decline in social function in patients with depression and has an important impact on the clinical prognosis of patients (MacQueen and Memedovich, 2017; Pan et al., 2019).

Among many cognitive functions, executive functions (EFs), as the core component of human intelligent activities, are considered as an advanced cognitive ability for individuals to control their thinking and behavior consciously and effectively. They are a series of goal-oriented cognitive control and adjustment processes, which can enable the individual's cognitive system to work effectively in a flexible and coordinated manner (Gilbert and Burgess, 2008), and it has an important impact on individual behavior and emotion (Diamond, 2013). Although the cognitive components of executive function have not been clearly defined, however, more and more researchers tend to agree that executive function includes three basic cognitive components: inhibition control, cognitive flexibility and working memory (Cristofori et al., 2019; Diamond, 2013; Schmitt et al., 2019). Previous studies have found that patients with depression have different aspects and degrees of impairment of executive function (Grahek et al., 2019; LeMoult and Gotlib, 2019), and the degree of executive function impairment was positively correlated with the severity of depressive symptoms (DeBattista, 2005; Ramos-Henderson et al., 2021). Patients with relatively good executive function also have better clinical outcomes after receiving antidepressant treatment (Wentao et al., 2020; Yan et al., 2015). Studies on the impairment of various components of executive function in patients with depression showed that patients had significant inhibition control, cognitive flexibility (Dotson et al., 2020; Palmwood et al., 2017) and working memory impairment (Dumas and Newhouse, 2015; Li et al., 2021a; Nikolin et al., 2021; Yoon et al., 2014). Moreover, the intervention training based on working memory could significantly alleviate the depressive symptoms of patients (Beloe and Derakshan, 2020; Chacko et al., 2014; Jopling et al., 2020; Nejati et al., 2020). In addition, relevant studies have found that depressive symptoms have a significant influence on complex cognitive field based on executive function, including financial capacity (Giannouli et al., 2022; Giannouli and Tsolaki, 2021) and decision-making (Hindmarch et al., 2013; Lawlor et al., 2020). Therefore, there is a close interaction between individual depressive symptoms and their executive function. However, previous studies have only evaluated the impairment of individual components of executive function and the predictive effect of overall performance of executive function on the prognosis of patients with depression. Currently, there is still no research that provides a more detailed and comprehensive assessment of executive function impairment in patients based on the three basic components of executive function, there has also been no in-depth analysis on whether the cognitive-behavioral performance of three components of executive function is associated with the degree of depression symptom relief after clinical treatment.

Therefore, the purpose of this study was to comprehensively assess the impairment of the three basic components of executive function

(inhibition control, cognitive flexibility and working memory) in patients with depression, and to further investigate whether the cognitive-behavioral performance of patients on these three components are associated with the degree of relief of depressive symptoms after clinical treatment. A systematic assessment of executive function impairment in patients with depression will help us gain a more comprehensive understanding of the characteristics of cognitive impairment in depression. Analyzing the relationship between executive function performance and the degree of relief of depressive symptoms after treatment in patients with depression can help us clarify behavioral markers for predicting clinical treatment outcomes in patients with depression. Hence, this study has important value for us to understand the pathological psychological mechanisms of depression and promote clinical interventions for depression.

Furthermore, previous study found that the degree of harmony in individual family relationships can affect their depressive state (Chen and Harris, 2019; Liu et al., 2017), poor family relationship status is an important factor that triggers individual depression, so this study intends to control for the degree of harmony in family relationships as an additional variable. In summary, based on previous research (Grahek et al., 2019; LeMoult and Gotlib, 2019; Wentao et al., 2020; Yan et al., 2015), we hypothesized that the performance of patients with depression in each component of executive function is significantly worse than that of healthy controls. Furthermore, the cognitive-behavioral performance of three components of executive function is associated with the degree of relief of depressive symptoms after clinical treatment.

## 2. Methods and materials

### 2.1. Participants

A total of 143 participants participated in this study, including 95 patients with depression (DEP) and 48 healthy controls (HC). DEPs were recruited from inpatients in a mental health hospital in Sichuan, China. HCs were recruited from the community where the hospital was located and matched with DEPs in terms of age, sex, and education level. Written informed consent was obtained from all participants. All procedures were approved by the Medical Ethics Committee of Guangyuan Mental Health Center (IRB No. GJWYL2019(1)-038).

All participants were evaluated and diagnosed by psychiatrists with >5 years of clinical diagnosis and treatment experience and the professional title of associate chief physician or above. The depression and anxiety levels of all participants were evaluated by professional psychological assessors with >5 years of clinical psychological assessment experience. The psychiatrists and psychological assessors who diagnosed and evaluated the participants were not aware of the content and purpose of this study.

The criteria for inclusion in the DEP group include: 1) met the ICD-10 diagnostic criteria for depression; 2) the standard score of Self-rating Depression Scale (SDS) is >53, and the total score of Hamilton Depression Scale (HAMD) is >17; 3) selective serotonin reuptake inhibitors (SSRIs) (e.g., Fluoxetine, Paroxetine, Citalopram, Fluvoxamine, Sertraline, and Escitalopram) and serotonin-norepinephrine reuptake inhibitors (SNRIs) (e.g., Venlafaxine, Duloxetine, and Minazepam) are only used for antidepressant treatment during and after hospitalization, and/or receive supportive psychotherapy during and after hospitalization. 4) age is between 18 and 55 years old, regardless of sex; 5) having primary school education level or above, enough to understand the requirements of experimental tasks; 6) normal vision or corrected vision, no color blindness; 7) good physical condition, no serious physical disease. The exclusion criteria include: 1) existing or previous brain injury, nervous system disease, dementia, or any other conscious change; 2) dependence or abuse of alcohol or other substances in the past 6 months; 3) existing or had suffered from schizophrenia, dissociative conversion disorder, eating disorders, and other serious mental diseases (not excluding patients with anxiety symptoms of depression); 4) other

conditions not suitable for participation in this study. The inclusion criteria of the HC group include that the standard score of the SDS of the individual is  $<50$ , and the total score of the HAMD is  $<7$ , after clinical psychological evaluation. The other inclusion criteria are the same as those of the DEP group except for items 1), 2) and 3). The exclusion criteria of the HC group are the same as those of the DEP, in addition to “having or having suffered from any mental disease in the ICD-10 diagnostic classification after assessment” and “having taken any neuro stimulants or inhibitors in the last 6 months”.

In this study, we conducted a baseline assessment of the depressive status of DEP before initiation of hospitalization and conducted a first follow-up assessment on the 15th day of hospitalization, and then conducted a second follow-up assessment approximately 9 months after discharge ( $M = 8.82$  months,  $SD = 0.86$ ). At the second follow-up, 3 DEPs were not evaluated because they could not get in touch (loss rate 3.16 %).

## 2.2. Depression assessment

The participants' depression level was assessed using the revised Chinese version of the Self Rating Depression Scale (SDS) (Wang and Chi, 1984; Zung and William, 1965) and the Chinese version of the Hamilton Depression Scale (HAMD-17) (Williams and Janet, 1988). The SDS was evaluated by participants, there were 20 items in total. The total score was 20 to 80. The higher the total score was, the more serious the depression was. The original score of the scale was multiplied by 1.25 to get the standard score. Previous studies have shown that the scale has good validity. In this study, the internal consistency reliability of the scale was 0.92. The HAMD-17 was evaluated by professional psychological evaluators, there were 17 items in total, with a total score of 0 to 53. The higher the total score, the more serious the depression. The internal consistency reliability of the scale in this study was 0.89.

In addition, we evaluated the harmony level of individual's family relationships using a question with a score of 1 to 5 (subjectively scored by the participants, with higher scores indicating more harmonious family relationships).

## 2.3. Execution functions assessment

The spatial-shape inhibition task was used to assess the inhibition control of participants. This study referred to the “Peach Blossom” task prepared by Davidson et al. (2006), used square and triangle graphics as stimulation materials. Participants were required to respond quickly and accurately to the keys according to the rules corresponding to the graphics and keys, regardless of whether the stimulus appears on the left or right side of the screen (for example, whether the triangle is on the left or right side of the screen, press the F key with the left hand). The task flow chart is shown in Fig. 1A. The task has 160 trials in total, half the stimulation position and key hand position are consistent, and half are inconsistent.

The N-back task was used to assess the working memory of participants (Nikolin et al., 2021). To make the task difficulty suitable for the participants in this study, the 1-back task was used to evaluate participants' ability to maintain working memory information. We used multi-angle geometry as stimulation materials. Participants were required to judge whether the previously presented geometry is completely consistent with the current one and to press the key quickly and accurately. The presentation process of stimulation is as follows: first present a fixation point, then present a geometric figure that needs to be memorized, then present a disturbing picture composed of four geometric figures, and then present a geometric figure for recognition and require participants to press the key to respond. The task flow chart is shown in Fig. 1B. The task has 120 trials in total, half the first and second geometric figures are consistent, and half are inconsistent.

The color-shape classification judgment task was used to assess the cognitive flexibility of participants. We referred to the dimension change card classification task prepared by Zelazo (2006), and used the red and blue geometric figures as stimulation materials. Participants were required to classify and judge the patterns according to the different dimensions of the stimulus (shape and color). In this task, participants were first required to classify the stimuli presented according to “shape”. After 8 trials, participants were required to classify the stimuli according to “color”. After 8 trials, they were also classified according to the “shape”. In this study, the participants were randomized according to

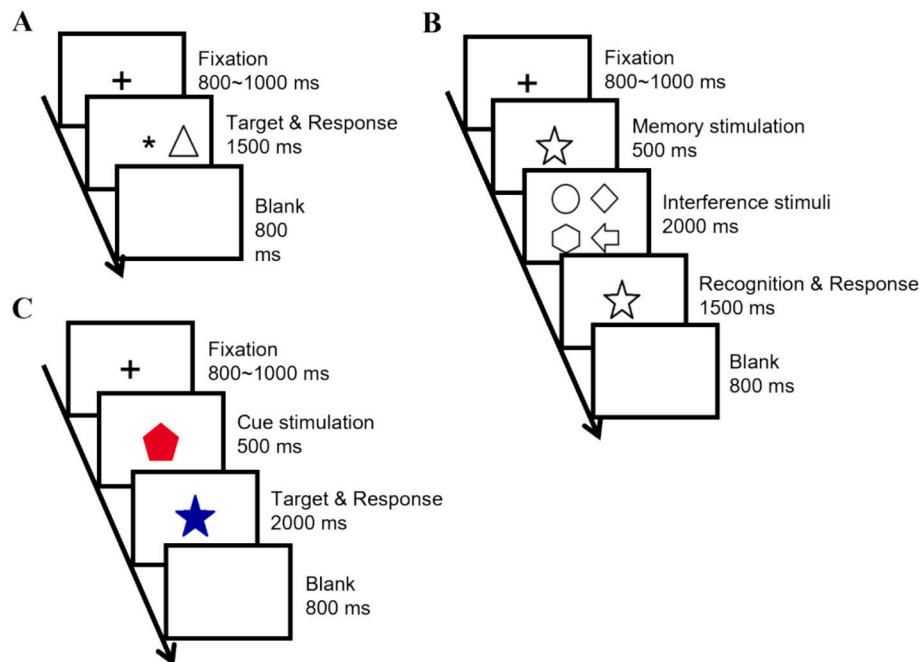


Fig. 1. Cognitive behavior assessment task flow chart.

Note. A) Spatial-shape inhibition task; B) One-back task; C) Color-shape classification judgment task. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the classification of “shape” or “color”, and “shape” to “color” and “color” to “shape” were converted five times. The task flow chart is shown in Fig. 1C. The task has 96 trials, half of which are classified by “shape” and half of which are classified by “color”, and half of which are consistent and inconsistent with the dimensions of the stimulus graphics and the reference graphics.

All the tasks were programmed with E-Prime 2.0 software, and the reaction times (RT) and accuracies of the participants were recorded.

#### 2.4. Procedure

All participants were assessed for demographic, medical history, and depressive status, and then completed cognitive behavior tests. In addition, we conducted the first follow-up assessment of the depressive status of the DEP on the 15th day of hospitalization, and the second follow-up assessment on the 8th to 9th month after their discharge.

The cognitive-behavioral tasks were conducted with Dell Vostro 5090 desktop computer, with 19.5-inch color display and screen resolution of 1024 × 768, the refresh rate is 60 Hz, the distance between the participant and the display is about 50 cm, and the screen background is white. All tasks were carried out in an independent and quiet laboratory. The test was conducted in order of “working memory – inhibition control – cognitive flexibility”, and the order was balanced between the participants. The whole test took about 30 min.

#### 2.5. Statistical analysis

The grand average RTs and accuracies of the 1-back task, and color-shape classification judgment task reflect the performance of participants' working memory and cognitive flexibility respectively. The difference of average RT and accuracy between the consistent and inconsistent trials in the spatial-shape inhibition task reflect the performance of participants' inhibition control. In addition, we used the baseline SDS and HAMD scale total scores of patients with depression to subtract the SDS and HAMD scale total scores from the first and second follow-up visits, respectively, to obtain the reduced SDS and HAMD scores after the first and second follow-up visits.

The independent-samples *t*-test was used to compare the difference between the HC group and DEP group in the scores of depression assessment, RTs and accuracies of cognitive behavior tasks. A multiple linear regression model was constructed with participants' age, sex, marital status, economic level, education level, medication status and harmony level of family relationships as covariates, RTs and accuracies of three components of cognitive-behavioral tasks as independent variables, and baseline SDS and HAMD scale total scores and reduced SDS and HAMD scores of DEP after two follow-up visits as dependent variables. Pearson correlation (2-tailed) was used to analyze the correlation between variables. In all analyses,  $p < 0.05$  indicates statistical significance. SPSS 25.0 is used for statistical analysis and processing of the collected data.

### 3. Results

#### 3.1. Demographic characteristics and depression assessment results

Table 1 shows the demographic characteristics of participants, the results of depression assessment at baseline and in the two follow-up visits, and the reduced depression scores after two follow-up visits. The statistical results of the anxiety levels of the DEP are shown in Table S1 of the supplementary materials. There were no significant differences in age, sex, marital status, economic level, and education level between the two groups. The harmony level of family relationships of DEP was significantly lower than those of the HC group,  $t(1,141) = 3.52, p = 0.001, d = 0.60$ . The total scores of SDS and HAMD of DEP in the baseline were significantly higher than those of the HC group (SDS scale total score:  $t(1,141) = 40.41, p < 0.001, d = 6.52$ ; HAMD total

**Table 1**

Demographic characteristics, depression assessment results, and the reduced depression scores after two follow-up visits.

| Study variables                                      | DEP (n = 95)  | HC (n = 48)   | $t/\chi^2$ | P-value |
|------------------------------------------------------|---------------|---------------|------------|---------|
| Age, years, $M \pm SD$                               | 37.07 ± 11.99 | 36.83 ± 11.79 | 0.11       | 0.91    |
| Sex, n (%)                                           |               |               | 0.004      | 0.95    |
| °Male                                                | 47 (49.5 %)   | 24 (50.0 %)   |            |         |
| °Female                                              | 48 (50.5 %)   | 24 (50.0 %)   |            |         |
| Marital status, n (%)                                |               |               | 0.67       | 0.71    |
| °Unmarried                                           | 22 (23.2 %)   | 13 (27.1 %)   |            |         |
| °Married                                             | 62 (65.2 %)   | 28 (58.3 %)   |            |         |
| °Divorce or bereavement                              | 11 (11.6 %)   | 7 (14.6 %)    |            |         |
| Economic level, n (%)                                |               |               | 1.02       | 0.60    |
| °Relatively poor                                     | 35 (36.8 %)   | 15 (31.3 %)   |            |         |
| °Secondary                                           | 43 (45.3 %)   | 26 (54.2 %)   |            |         |
| °Relatively rich                                     | 17 (17.9 %)   | 7 (14.6 %)    |            |         |
| Education level, n (%)                               |               |               | 0.22       | 0.98    |
| °Primary school                                      | 13 (13.7 %)   | 6 (12.5 %)    |            |         |
| °Junior high school                                  | 37 (38.9 %)   | 19 (39.6 %)   |            |         |
| °High school and technical secondary school          | 27 (28.4 %)   | 15 (31.3 %)   |            |         |
| °Undergraduate and junior college or above           | 18 (18.9 %)   | 8 (16.7 %)    |            |         |
| Harmony level of family relationships, $M \pm SD$    | 2.73 ± 0.90   | 3.33 ± 1.10   | 3.52       | 0.001   |
| Baseline assessment, $M \pm SD$                      |               |               |            |         |
| °SDS scale total score                               | 48.35 ± 3.06  | 32.15 ± 1.73  | 40.41      | <0.001  |
| °HAMD total score                                    | 23.50 ± 3.66  | 1.92 ± 1.11   | 52.93      | <0.001  |
| First follow-up assessment, $M \pm SD$               |               |               |            |         |
| °SDS scale total score                               | 36.53 ± 7.44  | –             | –          | –       |
| °HAMD total score                                    | 15.73 ± 4.54  | –             | –          | –       |
| Second follow-up assessment, $M \pm SD$              |               |               |            |         |
| °SDS scale total score                               | 33.35 ± 6.94  | –             | –          | –       |
| °HAMD total score                                    | 14.28 ± 4.71  | –             | –          | –       |
| Reduced score after the first follow-up, $M \pm SD$  |               |               |            |         |
| °SDS Scale Score                                     | 11.82 ± 6.16  | –             | –          | –       |
| °HAMD Score                                          | 7.77 ± 2.34   | –             | –          | –       |
| Reduced score after the second follow-up, $M \pm SD$ |               |               |            |         |
| °SDS Scale Score                                     | 15.00 ± 5.95  | –             | –          | –       |
| °HAMD Score                                          | 9.26 ± 2.64   | –             | –          | –       |

Note. DEP, patients with depression; HC, healthy controls; SD, standard deviation; SDS, self-rating depression scale; HAMD, Hamilton depression scale. P-values by ANOVA or Pearson Chi square tests (for linear association). Significant at  $p < 0.05$ . Second follow-up assessment  $n = 92$ .

score:  $t(1,141) = 52.93, p < 0.001, d = 7.98$ ).

#### 3.2. Cognitive-behavioral tasks results

The RTs and accuracies of three components of executive function in



the DEP group and HC group are shown in Fig. 2. The RTs of each components of executive function in the DEP group was significantly slower than that in the HC group,  $t(1,141) = 2.63\text{--}11.88$ ,  $ps < 0.01$ ,  $d = 0.51\text{--}1.91$ . Except for inhibition control, the accuracies of other components of executive function were significantly lower in the DEP group than in the HC group,  $t(1,141) = 2.60\text{--}3.32$ ,  $ps < 0.05$ ,  $d = 0.42\text{--}0.54$ . The results indicated that individuals with depression exhibit significant cognitive processing impairments in various components of executive function.

### 3.3. Correlation among three components of executive function and depression scores

The correlation coefficients among the three components of executive function and the HAMD score and SDS score are shown in Table 2 and Table S2, respectively. The results showed that, besides inhibition control RT and accuracy, the baseline and two follow-up visits HAMD and SDS scale scores were negatively correlated with the accuracies of other components of executive function, and positively correlated with the RTs. Moreover, the reduced HAMD and SDS scores after the first follow-up visits was significantly negatively correlated with the RTs of working memory and cognitive flexibility, respectively, and significantly positively correlated with the accuracy of cognitive flexibility. The reduced HAMD and SDS scores after the second follow-up visits were significantly negatively correlated with the RTs of working memory, respectively. The reduced SDS score after the second follow-up visits was significantly positively correlated with the accuracy of cognitive flexibility, and significantly negatively correlated with its RTs.

### 3.4. Results of multiple linear regression analyses

The results of multiple linear regression analysis using HAMD scores and SDS scores as outcome variables are shown in Table 3 and Table S3, respectively. When the demographic variables and harmony level of family relationships were covariates, there was a significant linear relationship between the RT and accuracy of working memory and the baseline HAMD and SDS scale total scores. When the reduced HAMD and SDS scores after the first and second follow-up visits were taken as dependent variables, respectively, only the RT of working memory had a significant linear relationship with them. The results showed that the RT of working memory had a stable linear relationship with the depressive state of the patients and had a significant association with the short-term

and long-term depression relief of patients after hospitalization. Moreover, the analysis results using two scale scores as outcome variables are similar, indicating consistency between the subjective assessment of depression symptoms by patients and the assessment of depression symptoms by physician. The linear regression effect of working memory on the baseline HAMD scores and reduced HAMD scores in different stages of DEP are shown in Fig. 3.

## 4. Discussion

The current study comprehensively assessed the performance of executive function in inpatients with depression and assessed their depressive status at baseline before hospitalization and at two follow-up visits after hospitalization. We systematically analyzed the impairment of the three basic components of executive function in DEP compared with HC and identified the association between the performance of each component of executive function in patients and the degree of relief of depressive symptoms after clinical treatment. The results showed that patients with depression perform worse in all components of executive function compared to healthy individuals. The performance of working memory and cognitive flexibility in DEP was significantly correlated with their depressive mood at baseline and at two follow-up visits. In addition, the RTs of patients' working memory were significantly correlated with the reduced depression scores after two follow-up visits. The results of the multiple linear regression found that only the RT of working memory had a significant linear relationship with the baseline depression scores and the reduced depression scores after two follow-up visits.

These findings are consistent with previous studies on impairment of executive function in patients with depression (Paelecke-Habermann et al., 2005; Roca et al., 2015; Vilgis et al., 2015), indicating that impairment of executive function is a major aspect of cognitive impairment in patients with depression. The results showed that baseline executive function was associated with depression severity and with improvement in depressive symptoms after treatment, this is consistent with Liu et al.'s study, which found a correlation between the improvements in executive function and the improvements of depressive symptoms in patients with depression after SSRI treatment (Liu et al., 2021). Furthermore, this study also found that the degree of impairment of working memory and cognitive flexibility was significantly positively correlated with the severity of depression, these results were consistent with previous studies (McDermott and Ebmeier, 2009; Roca et al.,

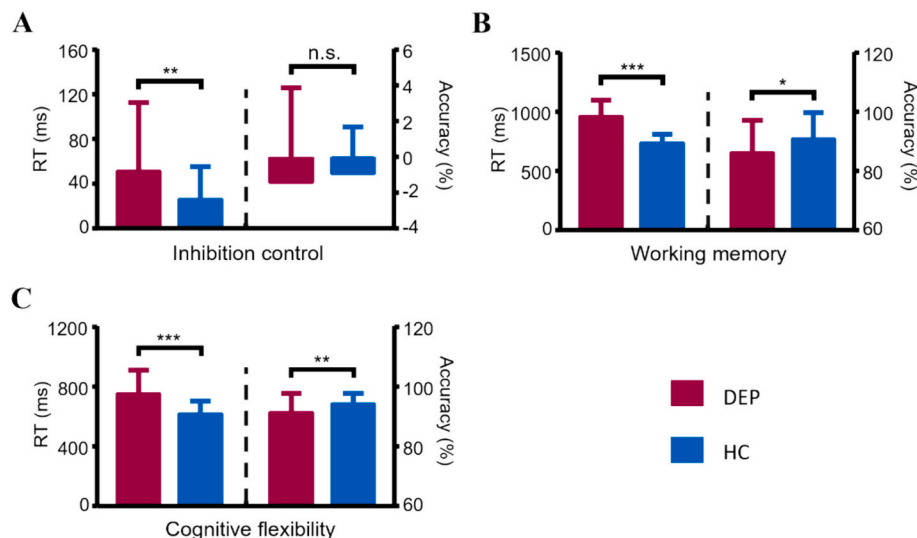


Fig. 2. RTs and accuracies of three components of executive function in the DEP group and HC group.

Note. The two data columns on the left of each subgraph are the RT, and the right is the accuracy. DEP, patients with depression; HC, healthy controls; RT, reaction times. \* $p < 0.05$ . \*\* $p < 0.01$ . \*\*\* $p < 0.001$ . n.s.,  $p > 0.05$ .

**Table 2**

Correlation coefficients for main study variables.

| Variables                                         | <i>n</i> | 1       | 2       | 3       | 4       | 5       | 6     | 7      | 8       | 9       | 10     |
|---------------------------------------------------|----------|---------|---------|---------|---------|---------|-------|--------|---------|---------|--------|
| 1. Accuracy of working memory                     | 143      | –       |         |         |         |         |       |        |         |         |        |
| 2. RT of working memory                           | 143      | –0.31** | –       |         |         |         |       |        |         |         |        |
| 3. Accuracy of cognitive flexibility              | 143      | 0.53**  | –0.46** | –       |         |         |       |        |         |         |        |
| 4. RT of cognitive flexibility                    | 143      | –0.36** | 0.74**  | –0.50** | –       |         |       |        |         |         |        |
| 5. Accuracy of inhibition control                 | 143      | 0.24**  | 0.04    | 0.21*   | –0.03   | –       |       |        |         |         |        |
| 6. RT of inhibition control                       | 143      | –0.08   | 0.01    | –0.03   | –0.01   | –0.32** | –     |        |         |         |        |
| 7. Baseline HAMD score                            | 95       | –0.34** | 0.74**  | –0.50** | 0.56**  | –0.01   | –0.09 | –      |         |         |        |
| 8. First follow-up HAMD scores                    | 95       | –0.38** | 0.80**  | –0.58** | 0.58**  | 0.05    | –0.15 | 0.86** | –       |         |        |
| 9. Second follow-up HAMD scores                   | 92       | –0.27*  | 0.77**  | –0.51** | 0.50**  | 0.05    | –0.10 | 0.83** | 0.94**  | –       |        |
| 10. Reduced HAMD score after the first follow-up  | 95       | 0.20    | –0.41** | 0.35**  | –0.26** | –0.12   | 0.15  | –0.11  | –0.60** | –0.52** | –      |
| 11. Reduced HAND score after the second follow-up | 92       | 0.02    | –0.37** | 0.20    | –0.12   | –0.12   | 0.12  | –0.11  | –0.50** | –0.65** | 0.80** |

Note. Pearson correlation coefficients *r* are presented. The HAMD scores of three times and reduced HAMD scores after follow-ups were only analyzed within the patients with depression group. RT, reaction times.

\* *p* < 0.05.

\*\* *p* < 0.01.

**Table 3**

Results of multiple linear regression analyses using HAMD scores as the outcome variable.

| Study variables                       | <i>R</i> <sup>2</sup> | <i>F</i> | <i>B</i> | <i>t</i> | 95 % CI         | VIF  |
|---------------------------------------|-----------------------|----------|----------|----------|-----------------|------|
| Model 1                               | 0.61                  | 16.92*** |          |          |                 |      |
| Age                                   |                       |          | 0.19     | 2.02*    | [0.001, 0.12]   | 1.99 |
| Sex                                   |                       |          | 0.161    | 2.35*    | [0.18, 2.16]    | 1.04 |
| Marital status                        |                       |          | –0.04    | –0.52    | [–1.28, 0.75]   | 1.45 |
| Economic level                        |                       |          | 0.04     | 0.52     | [–0.33, 0.55]   | 1.09 |
| Education level                       |                       |          | 0.14     | 1.58     | [–0.14, 1.19]   | 1.68 |
| Harmony level of family relationships |                       |          | –0.03    | –0.45    | [–0.71, 0.45]   | 1.14 |
| RT of working memory                  |                       |          | 0.67     | 8.25***  | [0.01, 0.02]    | 1.45 |
| Accuracy of working memory            |                       |          | –0.17    | –2.19*   | [–10.38, –0.50] | 1.28 |
| Model 2                               | 0.22                  | 3.56**   |          |          |                 |      |
| Age                                   |                       |          | –0.19    | –1.44    | [–0.09, 0.01]   | 1.97 |
| Sex                                   |                       |          | 0.14     | 1.42     | [–0.25, 1.52]   | 1.03 |
| Marital status                        |                       |          | 0.17     | 1.49     | [–0.23, 1.60]   | 1.45 |
| Economic level                        |                       |          | –0.03    | –0.30    | [–0.45, 0.33]   | 1.09 |
| Education level                       |                       |          | –0.01    | –0.11    | [–0.60, 0.54]   | 1.55 |
| Harmony level of family relationships |                       |          | 0.04     | 0.43     | [–0.40, 0.63]   | 1.13 |
| RT of working memory                  |                       |          | –0.36    | –3.16**  | [–0.01, –0.002] | 1.44 |
| Model 3                               | 0.18                  | 2.22*    |          |          |                 |      |
| Age                                   |                       |          | –0.19    | –1.38    | [–0.10, 0.02]   | 1.94 |
| Sex                                   |                       |          | 0.03     | 0.29     | [–0.91, 1.22]   | 1.05 |
| Marital status                        |                       |          | 0.10     | 0.85     | [–0.65, 1.62]   | 1.47 |
| Economic level                        |                       |          | –0.03    | –0.25    | [–0.53, 0.41]   | 1.10 |
| Education level                       |                       |          | –0.06    | –0.44    | [–0.83, 0.53]   | 1.57 |
| Harmony level of family relationships |                       |          | 0.01     | 0.12     | [–0.57, 0.64]   | 1.13 |
| Medication use                        |                       |          | –0.13    | –1.30    | [–1.83, 0.38]   | 1.04 |
| RT of working memory                  |                       |          | –0.32    | –2.69**  | [–0.01, –0.002] | 1.43 |

Note. Model 1 is a linear regression analysis of demographic variables, harmony level of family relationships, and executive function components on baseline HAMD scores (*n* = 95); Model 2 is a linear regression analysis of demographic variables, harmony level of family relationships, and executive function components on the reduced HAMD scores after the first follow-up (*n* = 95); Model 3 is a linear regression analysis of demographic variables, harmony level of family relationships, drug use, and executive function components on the reduced HAMD scores after the second follow-up (*n* = 92). The *R*<sup>2</sup> and *F* statistic are reported for the multiple linear regression models. RT, reaction times. 95 % CI, 95 % confidence interval. VIF, variance inflation factor.

\* *p* < 0.05.

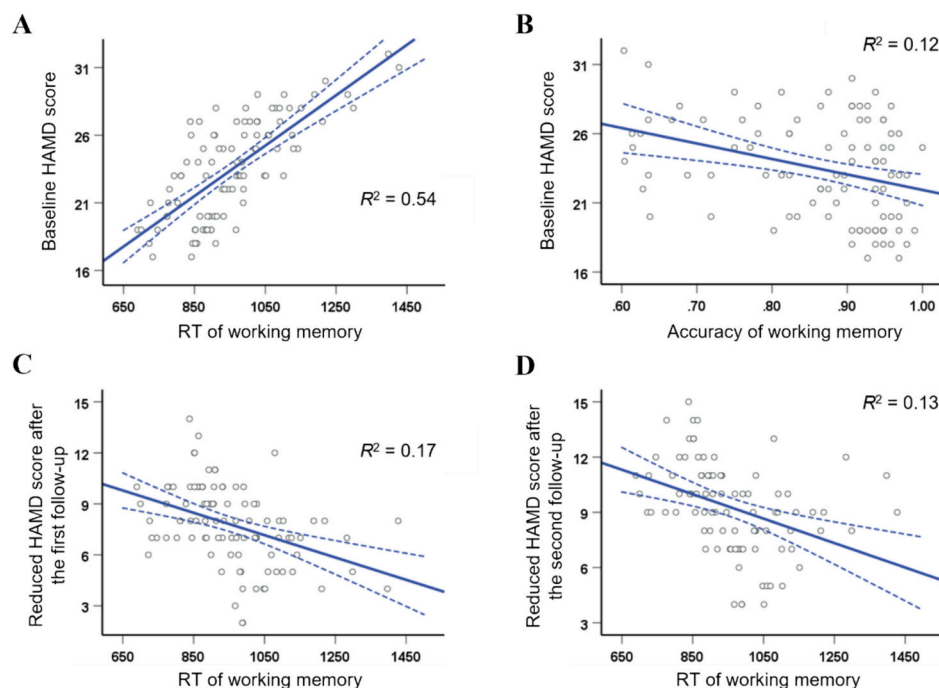
\*\* *p* < 0.01.

\*\*\* *p* < 0.001.

2015), which indicates a correlation between executive function impairment and depressive symptoms. However, we found no significant correlations between inhibition control among the three components of executive function and the level of depressive mood at baseline, as well as the degree of the reduced depression scores after two follow-up visits, indicating that the performance of inhibition control was not associated with their depressive status, which may be explained by the fact that the psychological function of inhibition control is mainly related to behavioral inhibition but not to emotional arousal and maintenance (Robertson et al., 1997), but deficits in inhibition control significantly increase impulsive behavior and suicide risk in patients with depression (Richard-Devantoy et al., 2015, 2016).

Most importantly, this study found that only the RT of working

memory had a significant linear relationship with the baseline depression scores and the reduced depression scores after two follow-up visits, which indicates that the RT of working memory could not only significantly predict the depressive state of patients with depression, but also has a significant association with the short-term and long-term depression relief of patients after hospitalization. Especially, the results of the linear regression model based on follow-up strongly demonstrate the special value of working memory in predicting the clinical prognosis of patients with depression. The reason may be that working memory, as an executive function component distinct from inhibition control and cognitive flexibility, is a capacity limited information processing system in which individuals temporarily store and process information during cognitive processes, it is the basis and core competence of general



**Fig. 3.** The linear regression effect of working memory on the baseline HAMD scores and the reduced HAMD scores in different stages of DEP.

*Note.* A) The linear regression effect of the RT of working memory on baseline HAMD score; B) The linear regression effect of the accuracy of working memory on baseline HAMD score; C) The linear regression effect of the RT of working memory on the reduced HAMD score after the first follow-up; D) The linear regression effect of the RT of working memory on the reduced HAMD score after the second follow-up. Dashed regions indicate 95 % confidence intervals. DEP, patients with depression; RT, reaction times.

cognitive function (Baddeley, 2010; Miller et al., 2018). Moreover, working memory can regulate an individual's emotional experience (Cohen and Kahana, 2022; Mikels and Reuter-Lorenz, 2019), by enhancing the working memory ability of patients with depression, it can make it easier for patients to remove negative information from working memory, thus reducing ruminant thinking and depressive mood (Jopling et al., 2020). Therefore, these results indicated that working memory is an important factor of depressive mood changes in patients and plays an important role in the generation, maintenance, and remission of depression.

In conclusion, the present study demonstrated that patients with depression have significant impairment of executive function, and that working memory is a critical factor in the generation, maintenance and remission of depression, and that patients' working memory performance can effectively predict their clinical prognosis. This is an important new discovery, which has great theoretical value for further understanding of the interaction between cognitive function and depression, as well as the cognitive psychological mechanism and development law of depression, in addition, it has important application value for clinical intervention and prognosis prediction of depression. By carrying out working memory training or regulating the neurophysiological basis related to working memory, enhance the working memory ability of patients with depression, thereby reducing the risk of aggravation of depression and increasing the possibility of relieving depression (Jopling et al., 2020). Furthermore, the results of this study have certain potential value for the clinical management of patients with depression. Accurately predict the clinical prognosis of patients with depression based on their working memory performance, and improve the adverse treatment prognosis by actively adjusting the treatment strategy, which will contribute to precision treatment and clinical individualized management of depression, and contribute to saving the treatment costs for the diseases and alleviating medical burden (Jones and Nemeroff, 2021).

## 5. Limitations and directions for future research

This study also has the following limitations. Firstly, this study only used cognitive behavior data as indicators to evaluate the cognitive performances of participants, the neurophysiological basis of impaired executive function in patients with depression has not been discussed. Secondly, this study only used cognitive-behavioral tasks that measure the three components of executive function, and the measured cognitive components are relatively small, so it is impossible to discuss the relationship between more cognitive performance and depression. Therefore, future studies could combine EEG, MRI, and other data to discuss the neurophysiological basis of the impairment of executive function in patients with depression, and further analyze the association between depression and cognitive processing by adding more cognitive components.

## 6. Conclusion

The patients with depression have significant executive function impairment, and their reaction time performance of working memory is associated with the degree of relief of depressive symptoms after clinical treatment.

## Funding statement

This work was supported by the National Natural Science Foundation of China (32171040, 32371105), the Sichuan Provincial Health Commission of Science and Technology Project (20PJ281), the Key R&D Projects of Science and Technology Plan of Science and Technology Department of Sichuan Provincial (2023YFS0085), the Sichuan Natural Science Foundation (Youth Fund) of Science and Technology Department of Sichuan Province (2024NSFSC1573), and the Postdoctoral Fellowship Program of CPSF (GZB20240658). Funders were not involved in the design and conduct of the study; collection,

management, analysis, and interpretation of the data; preparation, review or approval of the manuscript, and decision to submit the manuscript for publication.

## Ethical standards

The authors assert that all procedures contributing to this work comply with the ethical standards of the relevant national and institutional committees on human experimentation and with the Helsinki Declaration of 1975, as revised in 2008.

## CRediT authorship contribution statement

**Yongqing Hou:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Wen Liu:** Writing – review & editing, Investigation. **Tianbao He:** Writing – review & editing, Investigation, Data curation. **Antao Chen:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

## Declaration of competing interest

No authors of this paper have any conflicts of interest.

## Acknowledgements

The authors thank all participants in this study, we also thanks Ms. Hongni Xiang for her contribution in cognitive behavior data collection and Dr. Hong Liang, Dr. Xintong Deng, and Dr. Fan Liu for their assistance in the assessment of participants' clinical psychological status.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jad.2024.07.137>.

## References

- Baddeley, A., 2010. Working memory. *Curr. Biol.* 20, R136–R140. <https://doi.org/10.1016/j.cub.2009.12.014>.
- Beloe, P., Derakshan, N., 2020. Adaptive working memory training can reduce anxiety and depression vulnerability in adolescents. *Dev. Sci.* 23 <https://doi.org/10.1111/desc.12831>.
- Chacko, A., Bedard, A.C., Marks, D.J., Feisen, N., Uderman, J.Z., Chimiklis, A., Rajwan, E., Cornwell, M., Anderson, L., Ramon, M., 2014. A randomized clinical trial of Cogmed Working Memory Training in school-age children with ADHD: a replication in a diverse sample using a control condition. *J. Child Psychol. Psychiatry* 55, 247–255. <https://doi.org/10.1111/jcpp.12146>.
- Chen, P., Harris, K.M., 2019. Association of positive family relationships with mental health trajectories from adolescence to midlife. *JAMA Pediatr.* 173, e193336 <https://doi.org/10.1001/jamapediatrics.2019.3336>.
- Cohen, R.T., Kahana, M.J., 2022. A memory-based theory of emotional disorders. *Psychol. Rev.* 129, 742–776. <https://doi.org/10.1037/rev0000334>.
- COVID-19 Mental Disorders Collaborators, 2021. Global prevalence and burden of depressive and anxiety disorders in 204 countries and territories in 2020 due to the COVID-19 pandemic. *Lancet Lond. Engl.* 398, 1700–1712. [https://doi.org/10.1016/S0140-6736\(21\)02143-7](https://doi.org/10.1016/S0140-6736(21)02143-7).
- Cristofori, I., Cohen-Zimmerman, S., Grafman, J., 2019. Executive functions. *Handb. Clin. Neurol.* 163, 197–219. <https://doi.org/10.1016/B978-0-12-804281-6.00011-2>.
- Cummings, J.A.F., 2023. Nonsuicidal self-injury. *Am. J. Nurs.* 123, 44–47. <https://doi.org/10.1097/01.NAJ.0000938732.60723.b9>.
- Daly, M., Robinson, E., 2022. Depression and anxiety during COVID-19. *Lancet Lond. Engl.* 399, 518. [https://doi.org/10.1016/S0140-6736\(22\)00187-8](https://doi.org/10.1016/S0140-6736(22)00187-8).
- Davidson, M.C., Amso, D., Anderson, L.C., Diamond, A., 2006. Development of cognitive control and executive functions from 4 to 13 years: evidence from manipulations of memory, inhibition, and task switching. *Neuropsychologia* 44, 2037–2078. <https://doi.org/10.1016/j.neuropsychologia.2006.02.006>.
- DeBattista, C., 2005. Executive dysfunction in major depressive disorder. *Expert. Rev. Neurother.* 5, 79–83. <https://doi.org/10.1586/14737175.5.1.79>.
- Dehn, L.B., Beblo, T., 2019. Depressed, biased, forgetful: the interaction of emotional and cognitive dysfunctions in depression. *Neuropsychiatr. Klin. Diagn. Ther. Rehabil. Organ Ges. Österreichischer Nervenärzte Psychiatr* 33, 123–130. <https://doi.org/10.1007/s40211-019-0307-4>.
- Diamond, A., 2013. Executive functions. *Annu. Rev. Psychol.* 64, 135–168. <https://doi.org/10.1146/annurev-psych-113011-143750>.
- Dotson, V.M., McClintock, S.M., Verhaeghen, P., Kim, J.U., Draheim, A.A., Syzmkovic, S.M., Gradone, A.M., Bogoian, H.R., Wit, L.D., 2020. Depression and cognitive control across the lifespan: a systematic review and meta-analysis. *Neuropsychol. Rev.* 30, 461–476. <https://doi.org/10.1007/s11065-020-09436-6>.
- Dumas, J.A., Newhouse, P.A., 2015. Impaired working memory in geriatric depression: an fMRI study. *Am. J. Geriatr. Psychiatry Off. J. Am. Assoc. Geriatr. Psychiatry* 23, 433–436. <https://doi.org/10.1016/j.jagp.2014.09.011>.
- Giannouli, V., Tsolaki, M., 2021. Unraveling Ariadne's thread into the labyrinth of aMCI: depression and financial capacity. *Alzheimer Dis. Assoc. Disord.* 35, 363. <https://doi.org/10.1097/WAD.0000000000000417>.
- Giannouli, V., Stamovlasis, D., Tsolaki, M., 2022. Longitudinal study of depression on amnesic mild cognitive impairment and financial capacity. *Clin. Gerontol.* 45, 708–714. <https://doi.org/10.1080/07317115.2021.2017377>.
- Gilbert, S.J., Burgess, P.W., 2008. Executive function. *Curr. Biol.* 18, R110–R114. <https://doi.org/10.1016/j.cub.2007.12.014>.
- Grahek, I., Shenhav, A., Musslick, S., Krebs, R.M., Koster, E.H.W., 2019. Motivation and cognitive control in depression. *Neurosci. Biobehav. Rev.* 102, 371–381. <https://doi.org/10.1016/j.neubiorev.2019.04.011>.
- Hindmarch, T., Hotopf, M., Owen, G.S., 2013. Depression and decision-making capacity for treatment or research: a systematic review. *BMC Med. Ethics* 14, 54. <https://doi.org/10.1186/1472-6939-14-54>.
- Jones, C., Nemeroff, C.B., 2021. Precision psychiatry: biomarker-guided tailored therapy for effective treatment and prevention in major depression. *Adv. Exp. Med. Biol.* 1305, 535–563. [https://doi.org/10.1007/978-981-33-6044-0\\_27](https://doi.org/10.1007/978-981-33-6044-0_27).
- Jopling, E., Gotlib, I.H., LeMoult, J., 2020. Effects of working memory training on cognitive, affective, and biological responses to stress in major depression: a novel cognitive bias modification protocol. *J. Affect. Disord.* 265, 45–51. <https://doi.org/10.1016/j.jad.2020.01.007>.
- Knight, M.J., Baune, B.T., 2018. Cognitive dysfunction in major depressive disorder. *Curr. Opin. Psychiatry* 31, 26–31. <https://doi.org/10.1097/YCO.0000000000000378>.
- Lawlor, V.M., Webb, C.A., Wiecki, T.V., Frank, M.J., Trivedi, M., Pizzagalli, D.A., Dillon, D.G., 2020. Dissecting the impact of depression on decision-making. *Psychol. Med.* 50, 1613–1622. <https://doi.org/10.1017/S0033291719001570>.
- LeMoult, J., Gotlib, I.H., 2019. Depression: a cognitive perspective. *Clin. Psychol. Rev.* 69, 51–66. <https://doi.org/10.1016/j.cpr.2018.06.008>.
- Li, X., Chen, X.-L., Zhang, Y.-T., Li, R.-T., Bai, H.-P., Lui, S.S.Y., Chan, R.C.K., 2021a. Deficits in maintenance and interference control of working memory in major depression: evidence from the visuospatial change detection task. *Cogn. Neuropsychiatry* 26, 122–135. <https://doi.org/10.1080/13546805.2021.1871890>.
- Li, Y., Scherer, N., Felix, L., Kuper, H., 2021b. Prevalence of depression, anxiety and post-traumatic stress disorder in health care workers during the COVID-19 pandemic: a systematic review and meta-analysis. *PLoS One* 16, e0246454. <https://doi.org/10.1371/journal.pone.0246454>.
- Liu, Y., Fu, Q., Fu, X., 2009. The interaction between cognition and emotion. *Chin. Sci. Bull.* 54, 4102–4116. <https://doi.org/10.1007/s11434-009-0632-2>.
- Liu, J., Dong, X., Nguyen, D., Lai, D.W.L., 2017. Family relationships and depressive symptoms among Chinese older immigrants in the United States. *J. Gerontol. A Biol. Sci. Med. Sci.* 72, S113–S118. <https://doi.org/10.1093/gerona/glw138>.
- Liu, Q., He, H., Yang, J., Feng, X., Zhao, F., Lyu, J., 2020. Changes in the global burden of depression from 1990 to 2017: findings from the Global Burden of Disease study. *J. Psychiatr. Res.* 126, 134–140. <https://doi.org/10.1016/j.jpsychires.2019.08.002>.
- Liu, L., Lv, X., Zhou, S., Liu, Q., Wang, J., Tian, H., Zhang, K., Wei, J., Wang, C., Chen, Q., Zhu, G., Wang, X., Zhang, N., Huang, Y., Si, T., Yu, X., 2021. The effect of selective serotonin reuptake inhibitors on cognitive impairment in patients with depression: a prospective, multicenter, observational study. *J. Psychiatr. Res.* 141, 26–33. <https://doi.org/10.1016/j.jpsychires.2021.06.020>.
- MacQueen, G.M., Memedovich, K.A., 2017. Cognitive dysfunction in major depression and bipolar disorder: assessment and treatment options. *Psychiatry Clin. Neurosci.* 71, 18–27. <https://doi.org/10.1111/pcn.12463>.
- Malhi, G.S., Mann, J.J., 2018. Depression. *Lancet Lond. Engl.* 392, 2299–2312. [https://doi.org/10.1016/S0140-6736\(18\)31948-2](https://doi.org/10.1016/S0140-6736(18)31948-2).
- McDermott, L.M., Ebmeier, K.P., 2009. A meta-analysis of depression severity and cognitive function. *J. Affect. Disord.* 119, 1–8. <https://doi.org/10.1016/j.jad.2009.04.022>.
- Mikels, J.A., Reuter-Lorenz, P.A., 2019. Affective working memory: an integrative psychological construct. *Perspect. Psychol. Sci. J. Assoc. Psychol. Sci.* 14, 543–559. <https://doi.org/10.1177/1745691619837597>.
- Miller, E.K., Lundqvist, M., Bastos, A.M., 2018. Working memory 2.0. *Neuron* 100, 463–475. <https://doi.org/10.1016/j.neuron.2018.09.023>.
- Nejati, V., Salehinejad, M.A., Nitsche, M.A., Najian, A., Javadi, A.-H., 2020. Transcranial direct current stimulation improves executive dysfunctions in ADHD: implications for inhibitory control, interference control, working memory, and cognitive flexibility. *J. Atten. Disord.* 24, 1928–1943. <https://doi.org/10.1177/1087054717730611>.
- Nikolin, S., Tan, Y.Y., Schwaab, A., Moffa, A., Loo, C.K., Martin, D., 2021. An investigation of working memory deficits in depression using the n-back task: a systematic review and meta-analysis. *J. Affect. Disord.* 284, 1–8. <https://doi.org/10.1016/j.jad.2021.01.084>.
- Paelecke-Habermann, Y., Pohl, J., Lepow, B., 2005. Attention and executive functions in remitted major depression patients. *J. Affect. Disord.* 89, 125–135. <https://doi.org/10.1016/j.jad.2005.09.006>.



- Palmwood, E.N., Kropfing, J.W., Simons, R.F., 2017. Electrophysiological indicators of inhibitory control deficits in depression. *Biol. Psychol.* 130, 1–10. <https://doi.org/10.1016/j.biopsycho.2017.10.001>.
- Pan, Z., Park, C., Brietzke, E., Zuckerman, H., Rong, C., Mansur, R.B., Fus, D., Subramaniapillai, M., Lee, Y., McIntyre, R.S., 2019. Cognitive impairment in major depressive disorder. *CNS Spectr.* 24, 22–29. <https://doi.org/10.1017/S1092852918001207>.
- Ramos-Henderson, M., Ledezma-Dames, A., López, N., Machado Goyano Mac Kay, A.P., 2021. Executive functions and functional impairment in Latin seniors suffering from depression. *Neuropsychol. Dev. Cogn. B Aging Neuropsychol. Cogn.* 28, 543–558. <https://doi.org/10.1080/13825585.2020.1796915>.
- Richard-Devantoy, S., Szanto, K., Butters, M.A., Kalkus, J., Dombrovski, A.Y., 2015. Cognitive inhibition in older high-lethality suicide attempters. *Int. J. Geriatr. Psychiatry* 30, 274–283. <https://doi.org/10.1002/gps.4138>.
- Richard-Devantoy, S., Ding, Y., Lepage, M., Turecki, G., Jollant, F., 2016. Cognitive inhibition in depression and suicidal behavior: a neuroimaging study. *Psychol. Med.* 46, 933–944. <https://doi.org/10.1017/S0033291715002421>.
- Robertson, I.H., Manly, T., Andrade, J., Baddeley, B.T., Yiend, J., 1997. “Oops!”: performance correlates of everyday attentional failures in traumatic brain injured and normal subjects. *Neuropsychologia* 35, 747–758. [https://doi.org/10.1016/S0028-3932\(97\)00015-8](https://doi.org/10.1016/S0028-3932(97)00015-8).
- Roca, M., Vives, M., López-Navarro, E., García-Campayo, J., Gili, M., 2015. Cognitive impairments and depression: a critical review. *Actas Esp. Psiquiatr.* 43, 187–193.
- Rodríguez-Pascual, M., Álvarez-Subiela, X., Tor, J., Pardo, M., de la Serna, E., Sugranyes, G., Puig, O., Baeza, I., Dolz, M., 2021. Major depressive disorder and attenuated negative symptoms in a child and adolescent sample with psychosis risk syndrome: the CAPRIS study. *Eur. Child Adolesc. Psychiatry*. <https://doi.org/10.1007/s00787-021-01793-8>.
- Rudolph, K.D., Davis, M.M., Monti, J.D., 2017. Cognition-emotion interaction as a predictor of adolescent depressive symptoms. *Dev. Psychol.* 53, 2377–2383. <https://doi.org/10.1037/dev0000397>.
- Schmitt, L.M., Shaffer, R.C., Hessl, D., Erickson, C., 2019. Executive function in fragile X syndrome: a systematic review. *Brain Sci.* 9, E15. <https://doi.org/10.3390/brainsci9010015>.
- Vilgis, V., Silk, T.J., Vance, A., 2015. Executive function and attention in children and adolescents with depressive disorders: a systematic review. *Eur. Child Adolesc. Psychiatry* 24, 365–384. <https://doi.org/10.1007/s00787-015-0675-7>.
- Wang, Z., Chi, Y., 1984. Self-rating depression scale (SDS). *Shanghai Psychiatry* 71–72.
- Wentao, L., Minzhen, F., Canfang, Z., 2020. Effect of executive function on clinical efficacy in patients with depression. *Contemp. Med.* 26, 11–15. <https://doi.org/10.3969/j.issn.1009-4393.2020.35.005>.
- Williams, Janet, B.W., 1988. A structured interview guide for the Hamilton Depression Rating Scale. *Arch. Gen. Psychiatry* 45, 742–747. <https://doi.org/10.1037/t67131-000>.
- World Health Organization, 2017. *Depression and Other Common Mental Disorders: Global Health Estimates*. World Health Organization, Geneva.
- Yan, Y., Dong, Y., Jingping, Z., Jindong, C., Yihua, C., 2015. The cognitive function in treatment-resistant depression. *Chin. J. Clin. Psychol.* 23, 101–103. <https://doi.org/10.16128/j.cnki.1005-3611.2015.01.023>.
- Yoon, K.L., LeMoult, J., Joormann, J., 2014. Updating emotional content in working memory: a depression-specific deficit? *J. Behav. Ther. Exp. Psychiatry* 45, 368–374. <https://doi.org/10.1016/j.jbtep.2014.03.004>.
- Zelazo, P.D., 2006. The Dimensional Change Card Sort (DCCS): a method of assessing executive function in children. *Nat. Protoc.* 1, 297–301. <https://doi.org/10.1038/nprot.2006.46>.
- Zung, William, W.K., 1965. A self-rating depression scale. *Arch. Gen. Psychiatry* 12, 63–70. <https://doi.org/10.1001/archpsyc.1965.01720310065008>.